TECHNICAL DESIGN AND EVALUATION REPORT

First Signs

A Multilingual, Multi-Syndrome Screening and Closed-Loop Care-Navigation Platform for Bangladesh

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Requirements, Architecture, Implementation,
Feasibility, Comparative Study, and Repository Evaluation

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ABSTRACT

This report presents the requirements, architecture, implementation, feasibility, comparative position, and repository-level evaluation of *First Signs*, a multilingual, mobile-first screening and care-navigation prototype for Bangladesh. The platform supports five equally represented, non-diagnostic pathways: persistent cough or respiratory concern; fever or mosquito-borne illness concern; diarrhoea or dehydration concern; skin, sensation, or nerve concern; and child cough or difficult breathing. Its central contribution is a closed-loop workflow connecting consent, deterministic warning-sign screening, facility discovery, optional referral, independent attendance verification, missed-visit follow-up, pharmacy-scoped simulated incentives, and privacy-preserving public-health investigation signals.

Safety outcomes are generated by fixed, versioned TypeScript rules. Optional artificial-intelligence and acoustic services are prevented by design from overriding those outcomes. The implementation uses Next.js 16, React 19, strict TypeScript, Supabase/PostgreSQL, MapLibre, QR tooling, and optional Gemini, Telegram, Resend, and Google HeAR integrations. Repository evaluation performed on 19 October 2026 found that linting, TypeScript checking, and the production build completed successfully; 38 focused automated test cases also passed. These findings establish software consistency in the evaluated scope, not clinical effectiveness. The prototype is technically feasible for continued research, while deployment remains contingent on expert review, verified multi-pathway facility data, accessibility and usability studies, operational partnerships, privacy governance, security assessment, and prospective field evaluation.

Index Terms—digital health, care navigation, deterministic triage, multilingual health systems, Bangladesh, referral workflow, privacy, pharmacy engagement, public health, software engineering.

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I. Introduction

A. Project Overview

People can be lost at every handoff between noticing a warning sign and reaching formal care: interpreting symptoms, finding an appropriate service, preserving a referral, attending the visit, and recovering after a missed visit. Bangladesh research reports delayed care seeking and frequent initial contact with nearby pharmacies among people with tuberculosis symptoms [1]. Digital symptom-checker studies also report variable triage and diagnostic performance [2], [3], [4]. These findings motivate a system that is explicit about its limits and optimized for safe navigation rather than diagnosis.

*First Signs*addresses this discontinuity through a closed loop:

Concern → safe guidance → facility → referral → attendance → follow-up.

The architecture is multilingual; the present implementation provides complete English and Bangla interfaces. All five pathways are first-class entry points. Respiratory audio research and the current TB-evaluation facility directory are bounded components, not the identity of the platform.

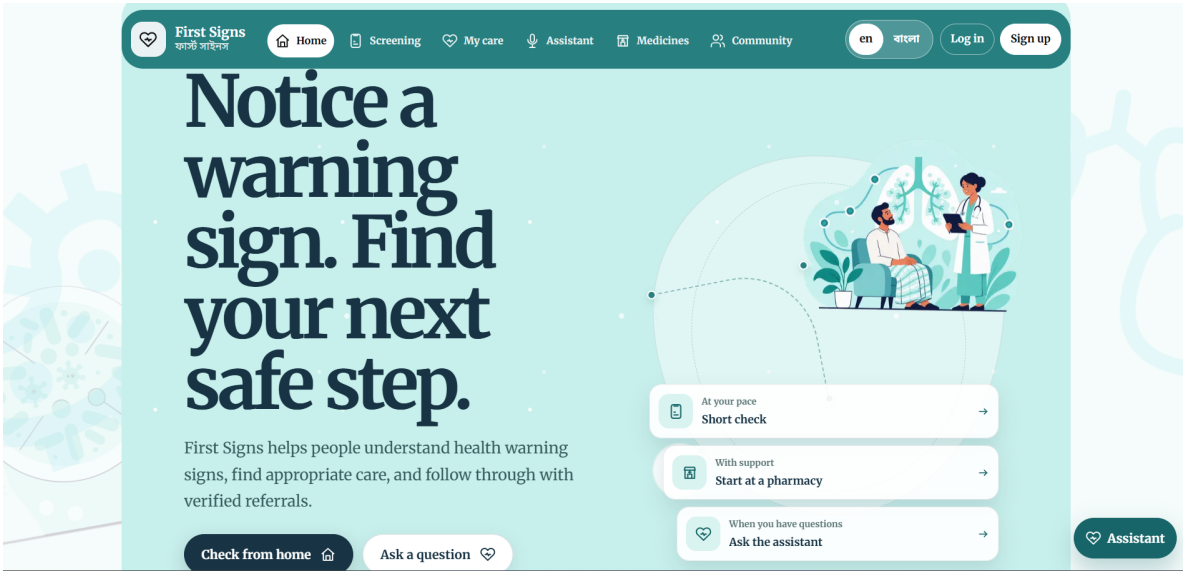


Fig. 1. First Signs landing page with language choice and separated self-use and assisted entry.

B. Problem Statement

The target problem is not simply insufficient health information. It is the loss of continuity between early concern and completed formal evaluation. A standalone symptom questionnaire may produce advice without a destination; a verbal recommendation may be forgotten; a missed appointment may end the journey; and an aggregate dashboard may expose more patient data than programme teams need. The engineering problem is to connect these transitions while preserving safety, patient choice, role separation, and minimum-data principles.

C. Objectives and Scope

The objectives are to describe the multi-pathway requirements and implementation; document deterministic safety, privacy, incentive, and failure boundaries; compare the design with related research and common system archetypes; report repository experiments without overstating them; and identify evidence needed for responsible continuation. The scope excludes diagnosis, prescribing, clinical validation, epidemiological claims, regulatory approval, deployment economics, and real-world effectiveness claims.

D. Main Tools and Technology Stack

TABLE I
IMPLEMENTED TECHNOLOGY STACK AND RESPONSIBILITY.

Layer	Technology	Responsibility
Web	Next.js 16.3.8, React 19.2.8	Pages, server rendering, route handlers, interactive workflows
Language	TypeScript 5.9 strict mode	Shared types, validation, deterministic care rules
Data and identity	Supabase, PostgreSQL, `@supabase/ssr`	Auth, persistence, RLS, transactional functions, scoped aggregates
Maps and motion	MapLibre GL JS 6.9; Motion 14.3	Facility maps, routes, broad-area views, restrained transitions
Referrals	`qrcode`, `jsqr`	QR generation and local QR-image parsing
Optional services	Gemini, Telegram, Resend, HeAR	Explanation, messaging, reminders, transient acoustic research
Testing	Node test runner, Playwright Core	Unit, integration, browser, language, responsive checks

E. Proposed and Implemented Feature Comparison

TABLE II
FEATURE STATUS AT REPORT PREPARATION.

Capability	Status	Evidence or remaining gate
Five deterministic care pathways	Implemented	Shared rules, screens, validation, and tests
English and Bangla experience	Implemented	Persistent toggle; independent language review remains required
Facility discovery and routes	Implemented	Broad-division data and optional transient GPS
Referral ID, token, and QR	Implemented	Transactional creation and private ticket access
Independent attendance verification	Implemented	Intended-facility checks and one-time completion
Missed-visit recovery	Implemented	Barrier-specific next actions in My Care
Pharmacy metrics and credits	Implemented	Simulated credits only for assisted journeys
Grouped public-health signals	Implemented	Investigation signals, not outbreak claims
AI, email, messaging, and audio	Conditional	Configuration-dependent with safe-degradation boundaries
Clinically approved content	Future work	Qualified review and prospective validation required
Nationwide verified directory	Future work	Present service coverage is incomplete and partly TB-centred
Production governance	Future work	Prototype controls are insufficient for deployment

II. RESEARCH BACKGROUND, REQUIREMENTS, AND DESIGN

A. Related Works

Digital-triage research establishes a cautious baseline. *Evaluation of Symptom Checkers for Self Diagnosis and Triage: Audit Study* reported substantial variation across public systems [2]. *The Diagnostic and Triage Accuracy of Digital and Online Symptom Checker Tools: A Systematic Review* [3] and *Triage and Diagnostic Accuracy of Online Symptom Checkers: Systematic Review* [4] likewise found variable performance and limited evidence for safe broad adoption. This supports small, inspectable rules for care-action categories while keeping generative AI outside safety decisions.

WHO states that digital interventions supplement rather than replace health systems [5]; WHO SMART Guidelines emphasize standards-based, machine-readable, adaptive, requirements-based, and testable specifications [6]. *First Signs*follows that direction conceptually through language-neutral outcomes and versioned rules without claiming formal conformance.

TABLE III: Research-paper titles and relevance to the five-pathway design.

Area	Research-paper title	Design relevance
Respiratory/TB	*Mixed-methods Study to Assess Delay among Patients with Tuberculosis in an Urban Setting of Bangladesh* [1]	Care delay and Bangladesh context
Pharmacy referral	*Can Community Pharmacists Improve Tuberculosis Case Finding? A Mixed Methods Intervention Study in India* [7]	Assisted referral and incentives
Navigation	*Can Patient Navigators Help Potential TB Patients Navigate the Diagnostic and Treatment Pathways? An Implementation Research from India* [8]	Continuity across care journeys
Fever/dengue	*District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning* [9]	Predictive surveillance contrast
Fever/malaria	*Knowledge on the Transmission, Prevention and Treatment of Malaria among Two Endemic Populations of Bangladesh and Their Health-Seeking Behaviour* [10]	Knowledge and delayed care seeking
Diarrhoea	*Effects of a Water, Sanitation, and Hygiene Mobile Health Program on Diarrhea and Child Growth in Bangladesh: A Cluster-Randomized Controlled Trial of the CHoBI7 Mobile Health Program* [11]	Mobile follow-up in Bangladesh
Skin/leprosy	*Leprosy Screening Based on Artificial Intelligence: Development of a Cross-Platform App* [12]	Mobile research; visual diagnosis avoided here
Assisted digital health	*Efficacy of Using a Digital Health Intervention Model Using Community Health Workers for Primary Health Services in Bangladesh: A Repeated Cross-Sectional Observational Study* [13]	Assisted delivery context
Health acoustics	*HeAR—Health Acoustic Representations* [14]	Optional embeddings, not diagnosis

B. Comparative Study

The distinguishing unit is the closed loop, not a novel diagnostic model. This is an architectural comparison, not a controlled product trial.

TABLE IV
CONCEPTUAL COMPARISON WITH COMMON DIGITAL-HEALTH ARCHETYPES.

Capability	Symptom checker	Health chatbot	Referral registry	Surveillance view	*First Signs*
Warning-sign guidance	Sometimes	Sometimes	No	No	Yes
Deterministic urgent precedence	Variable	Variable	No	No	Yes
Multiple patient pathways	Usually	Usually	Limited	Aggregate only	Five
Portable referral and QR	Rare	Rare	Yes	No	Yes
Independent attendance confirmation	Rare	Rare	Yes	No	Yes
Missed-visit recovery	Limited	Limited	Sometimes	No	Yes
Pharmacy/self-use separation	Rare	Rare	Sometimes	No	Yes
Privacy-safe grouped signals	Rare	Rare	Sometimes	Yes	Yes
Safe optional-service degradation	Variable	Variable	N/A	N/A	Required

C. Requirement Analysis

Functional requirements include two entry modes; consent; five pathway-specific screens; urgent precedence; language-neutral results; facility discovery; optional referral; referral ID, token, and QR; independent attendance confirmation; private care history; missed-visit support; pharmacy-scoped metrics; reward safeguards; anonymous community feedback; and grouped programme signals.

Non-functional requirements include mobile responsiveness, readable health content, equivalent English/Bangla meaning, server-side validation, minimum-data collection, scoped authorization, idempotent state changes, graceful degradation, secret isolation, and regression-testability. Merriweather is required for English and Noto Sans Bengali for Bangla.

TABLE V
STAKEHOLDER CAPABILITIES AND ACCESS BOUNDARIES.

Stakeholder	Required capability	Boundary
Person	Screen, receive guidance, choose referral, review own journey	No other person's record; referral remains optional
Pharmacy	Assist linked journey; view own aggregates and credits	No self-use records, diagnosis, or attendance approval
Facility staff	Lookup intended referral and confirm attendance once	No pharmacy reward control
Public health	View grouped metrics and broad-area signals	No identity, raw answers, GPS, token, or audio
Optional services	Explain, route, message, remind, or process audio	No safety authority

D. Methodology

Development follows a requirements-driven, safety-first method: define pathway danger signs and outputs; encode language-neutral rules with urgent precedence; separate translated presentation from calculation; validate at client and server boundaries; separate private records from minimum signals; model referral, verification, follow-up, and rewards as controlled transitions; isolate optional integrations; and connect critical boundaries to tests. Explainability is treated as inspectability: a reviewer can locate a rule version, validation, outcome code, display copy, persistence function, and regression test.

E. Closed-Loop Care Design

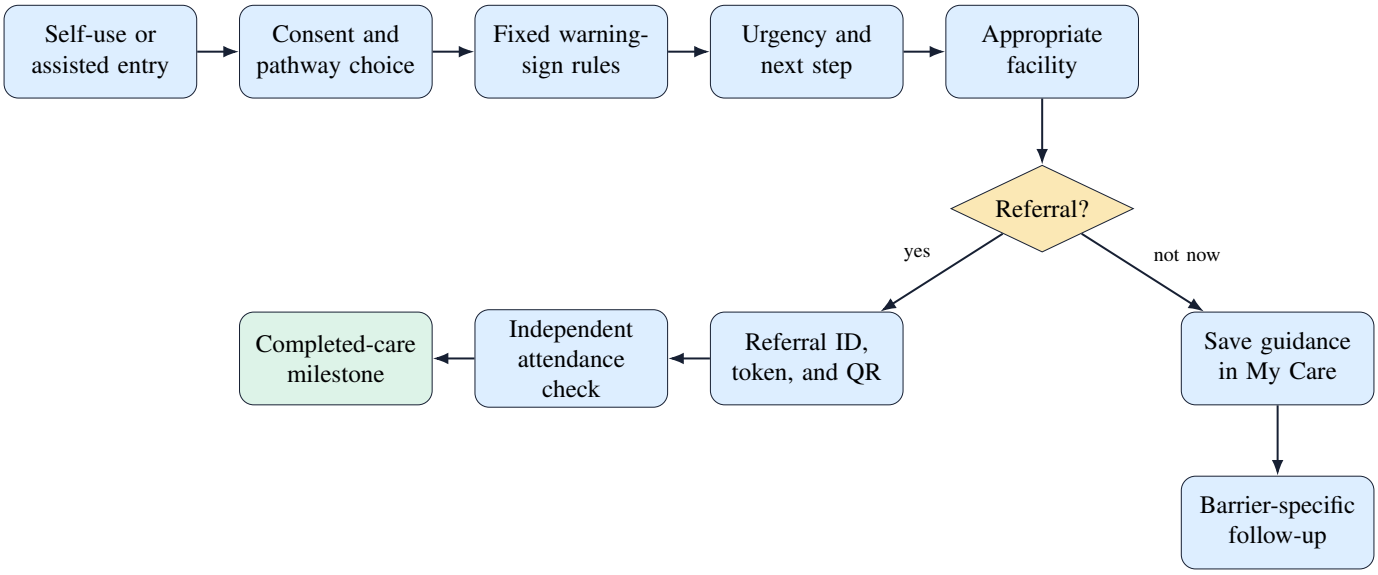


Fig. 2. Closed-loop care journey with separated forward and recovery paths.

F. System Architecture

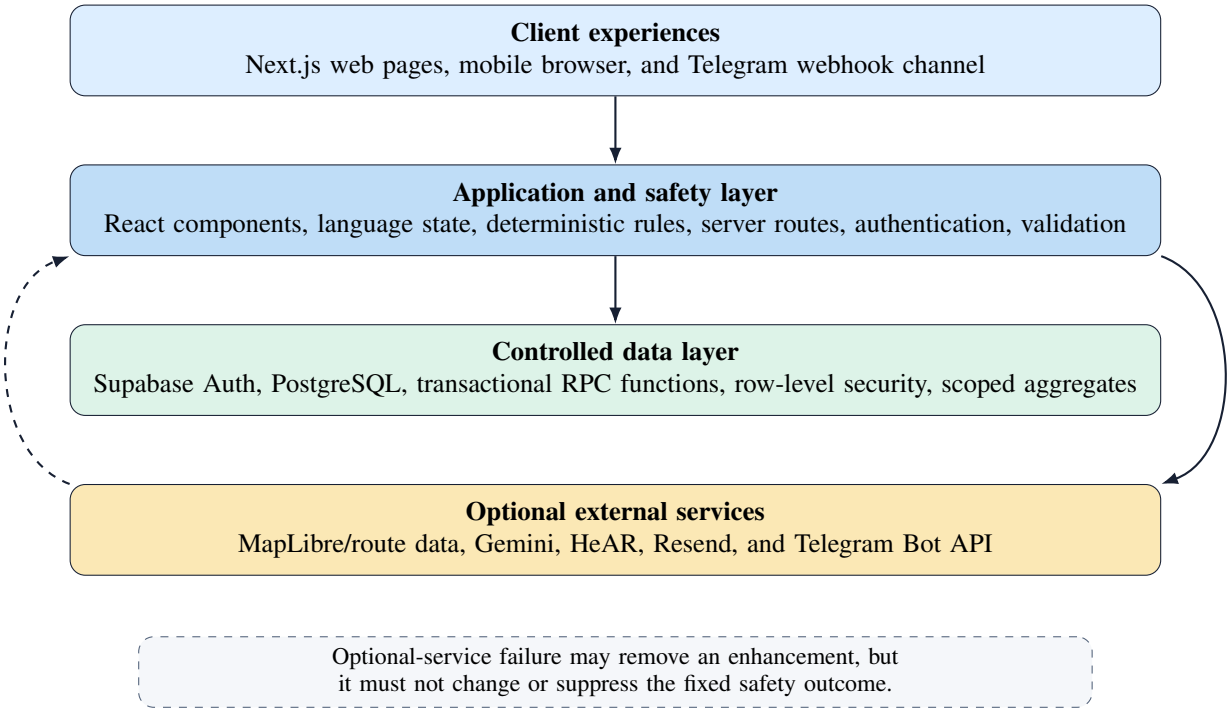


Fig. 3. Layered technical architecture and optional-service boundary.

G. Deterministic Decision Design

Each pathway maps validated inputs to one of three language-neutral outcomes:

$$O \in \{\texttt{CONTINUE_WITH_CARE_INFORMATION}, \texttt{PROFESSIONAL_EVALUATION_RECOMMENDED}, \texttt{URGENT_MEDICAL_EVALUATION}\}.$$

Urgent danger signs are evaluated first. Fever rules do not guess dengue versus malaria; skin/nerve rules do not diagnose leprosy; and child-breathing rules do not label pneumonia.

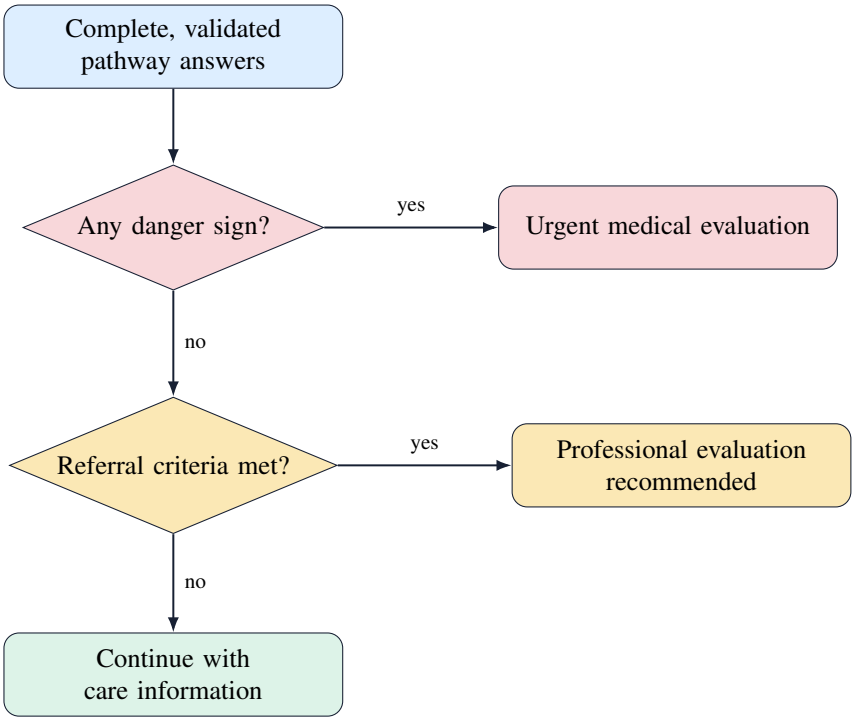


Fig. 4. Shared outcome precedence across all pathways.

H. Data, Privacy, and Incentive Design

Private care state and anonymous programme signals are different concepts. Exact GPS may be transiently used for a route but is not persisted in screening, referral, analytics, or hotspot records. Raw audio and embeddings are transient. Telegram continuity uses a one-way identifier; reminder emails are generic.

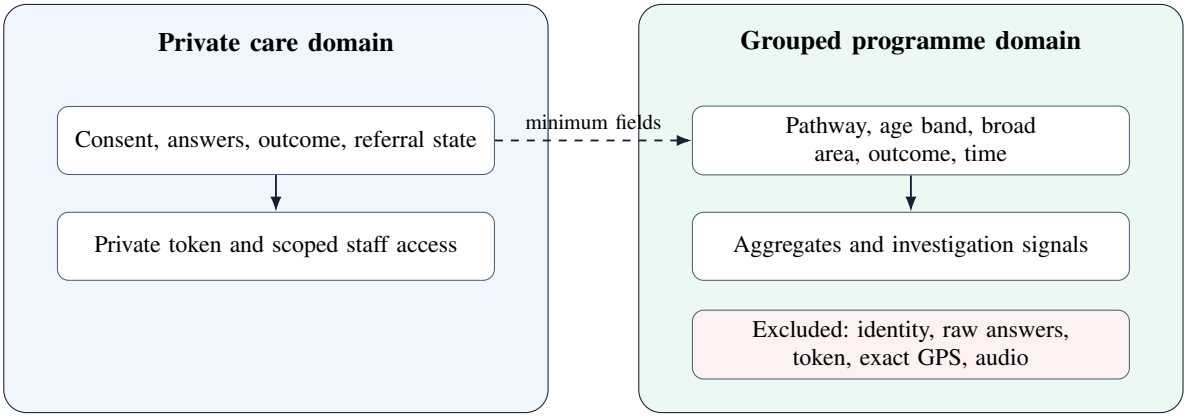
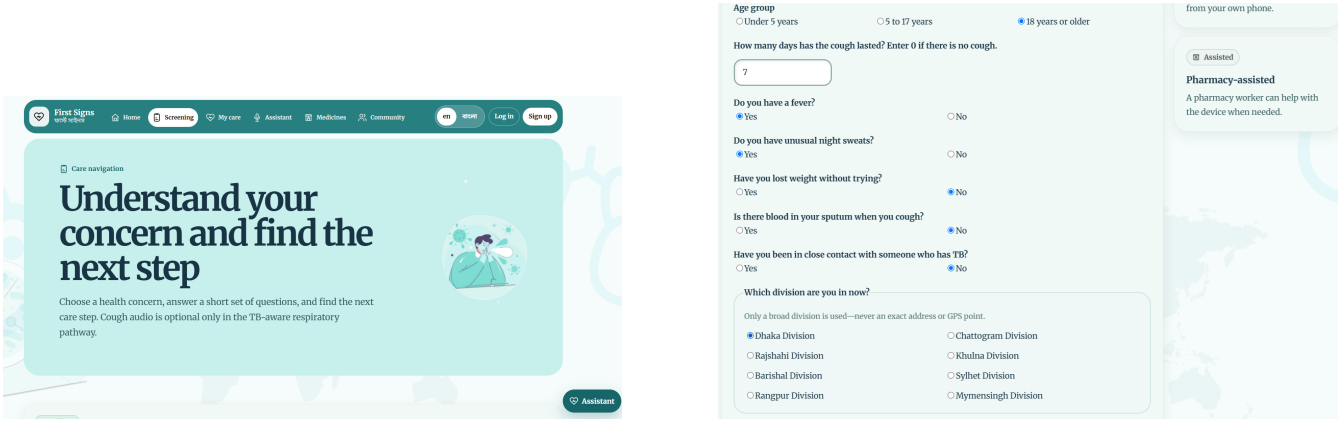


Fig. 5. Privacy boundary between individual care and grouped signals.

Pharmacy rewards remain separate from clinical outcomes. Self-use journeys cannot create pharmacy rewards; pharmacies cannot verify their own referrals; completion is idempotent; and unusual activity may hold a simulated reward but cannot block care.

I. Multilingual and Mobile-First Interface Design



(a) Screening introduction (b) Structured pathway questions

Fig. 6. Updated responsive screening workflow. English is shown; the same interface is available in Bangla through the persistent language control.

III. Implementation and Software Engineering Work

A. Program Structure

At analysis time, the scoped directories contained 56 TypeScript files, 38 TSX files, 43 SQL files, 21 MJS files, 8 MTS files, and 9 Python files. The source included 13 user-facing pages, 21 API/callback routes, 33 ordered Supabase migrations, and 36 test/check artifacts. Counts describe scale, not quality.

Core modules include `lib/care-pathways.ts`, `components/screening-flow.tsx`, `lib/care-session.ts`, `lib/auth.ts`, `lib/supabase-server.ts`, `lib/health-rag.ts`, and `lib/messaging-server.ts`.

TABLE VI
Representative requirement-to-evidence traceability.

Requirement	Implementation evidence	Verification evidence
Five deterministic pathways	`lib/care-pathways.ts`	Care-pathway suite
Respiratory urgent precedence	`lib/safety-rules.ts`	Safety-rule suite
Server input validation	Screening API route	Pathway/integration checks
Transactional referral IDs	Referral-counter migration	SQL invariant test
Independent verification	Referral verification API	Verification-flow check
Grouped public-health view	Dashboard API and library	Dashboard/privacy checks
AI claim boundary	`lib/ai-companion.ts`	AI guardrail suite
Transient audio hash	`lib/audio-hash.ts`	Audio-hash suite

B. Request and State-Transition Implementation

The client records self-use or validated-pharmacy entry. After consent and complete answers, shared rules calculate a language-neutral result. The server independently validates and recalculates before invoking a controlled database function. Referral creation generates a display identifier and opaque token.

Verification checks authorization, intended facility, state, and prior use. Pharmacy credit eligibility is considered only after an independent milestone.

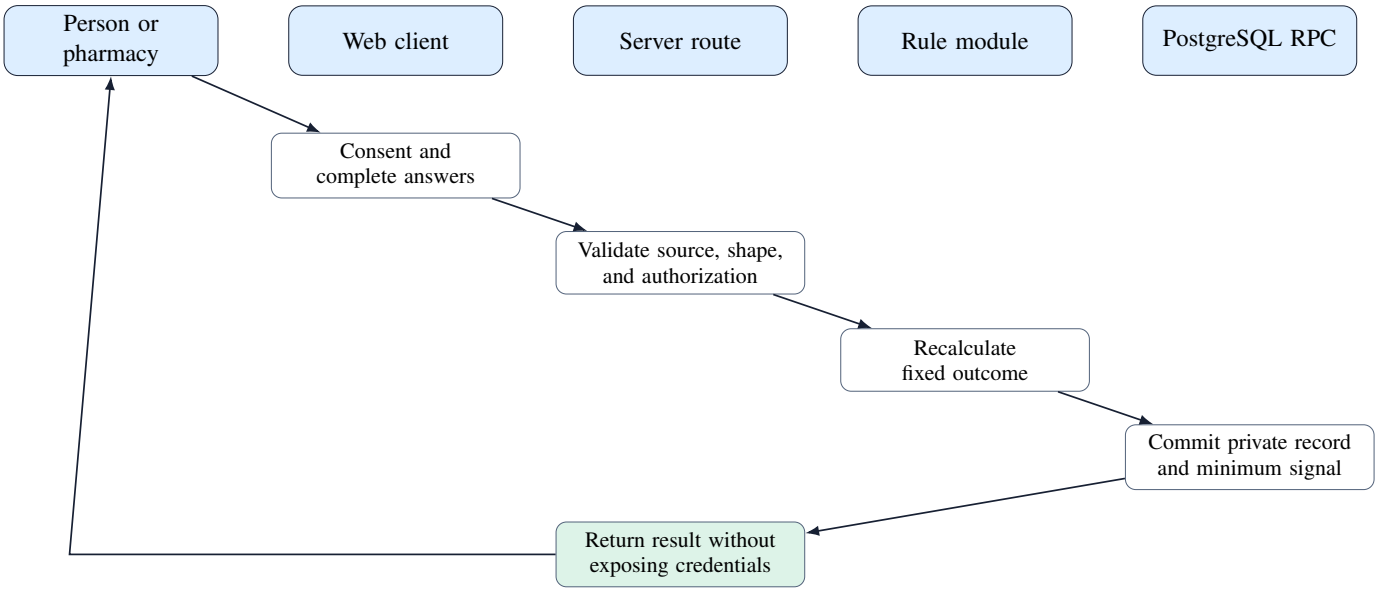
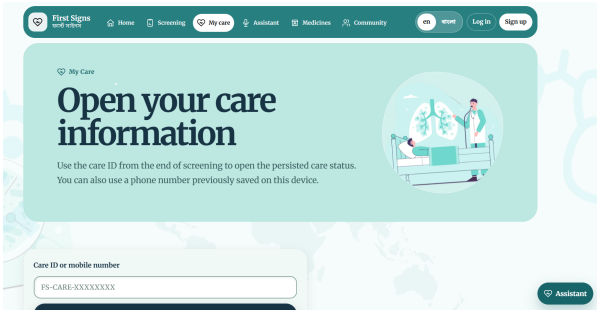


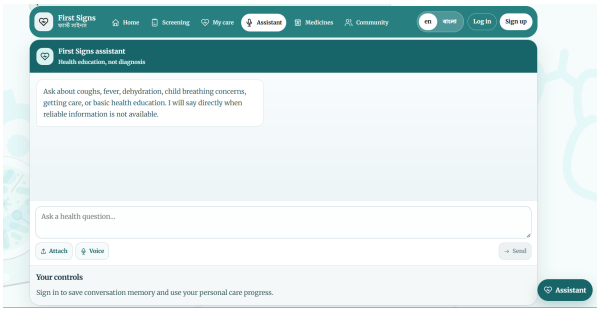
Fig. 7. Screening request sequence with server-side recalculation.

C. Referral, Verification, and Follow-Up

Referral creation is optional and transactional. A public care identifier supports communication while an opaque token protects private reopening. Attendance means formal evaluation started; it is not a diagnosis, test result, or treatment outcome. Missed visits record structured barriers and return practical next actions.



(a) Private My Care continuity

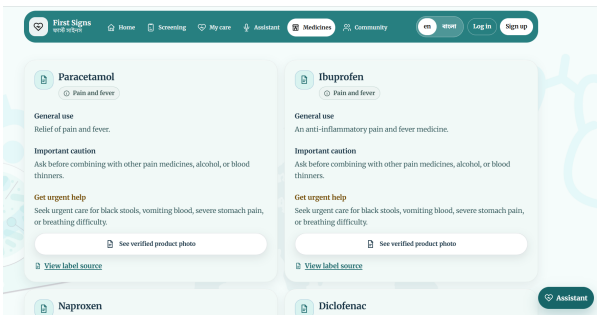


(b) Bounded health assistant

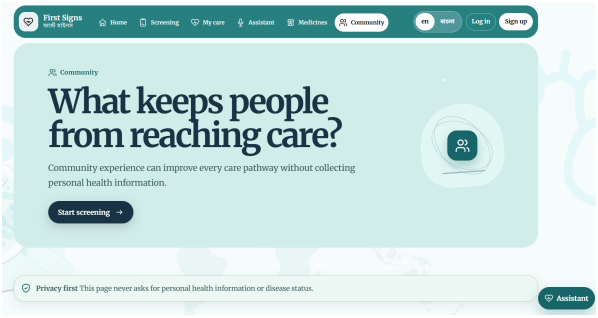
Fig. 8. Care continuity and education; the assistant cannot alter safety outcomes.

D. Education and Community Participation

Medicine Corner provides general information without personal dosing or prescribing. Community feedback accepts structured barriers without names, phone numbers, free text, or disease status.



(a) General medicine education



(b) Anonymous barriers-to-care feedback

Fig. 9. Non-diagnostic education and privacy-minimized participation.

E. Authentication, Transactions, and Integrity

Supabase Auth provides sessions; server routes resolve role and scope. RLS protects exposed tables and narrowly granted functions perform controlled changes. Privileged keys remain server-side. Transactional counters prevent rolled-back or retried requests from consuming referral numbers. Repeated rewards, audio hashes, referral reuse, and unusual activity lead to narrow review states, never denial of care.

F. Optional Services and Failure Engineering

If Gemini is unavailable, screening and referral continue. If HeAR decoding, inference, or connectivity fails, symptom-based care continues. If route drawing fails, facility details remain available. If reminder delivery fails, referral state does not change. Telegram follows the same server-owned rules.

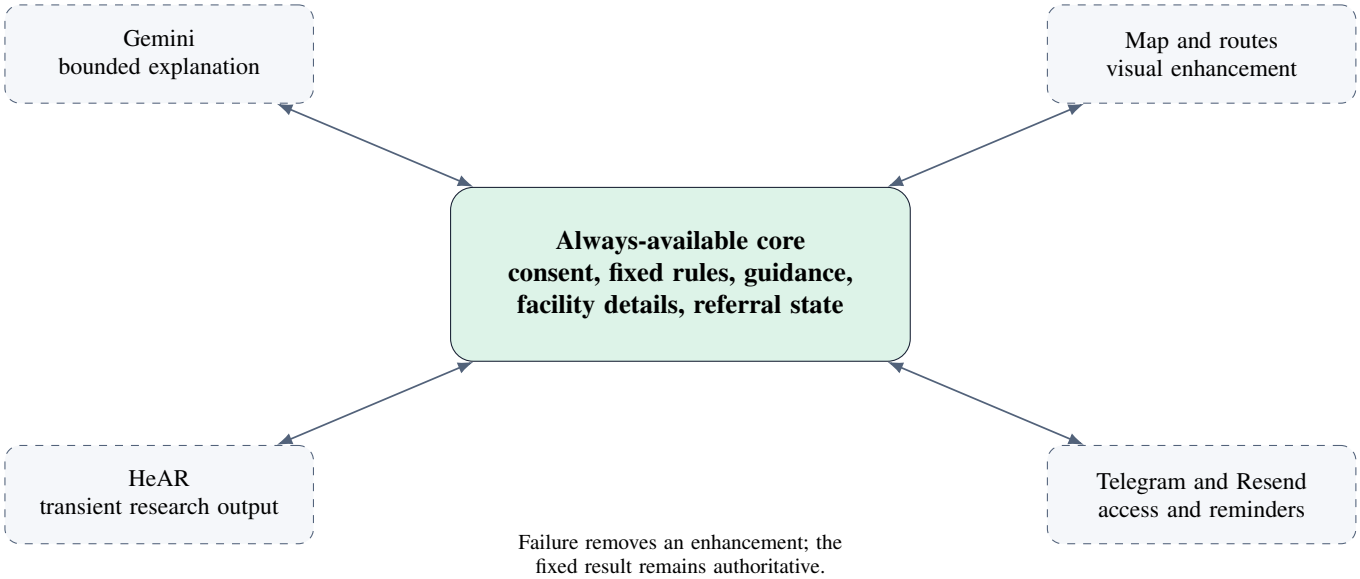


Fig. 10. Graceful degradation for optional integrations.

IV. Results, Feasibility, and Discussion

A. Evaluation Protocol

Repository checks were executed on 19 October 2026. Windows command shims were used because local PowerShell policy blocked the `npm.ps1` and `npx.ps1` wrappers. Evaluated commands were `npm.cmd run lint`, `npx.cmd tsc -noEmit`, `npm.cmd run build`, and focused suites for pathways, respiratory safety, configuration, audio hashing, AI guardrails, medicine data, and messaging security.

B. Results and Experiments

ESLint and TypeScript checks completed successfully. The Next.js 16.3.8 production build compiled, type-checked, generated static output, and listed expected routes. Seven suites executed 38 tests; all passed with none failed, cancelled, skipped, or todo.

TABLE VII
FOCUSED AUTOMATED RESULTS OBSERVED DURING REPORT PREPARATION.

Suite	Passed	Failed	Evaluated scope
Care pathways	6	0	Five paths and adult child-path boundary
Respiratory safety	9	0	Duration, symptoms, urgency, invalid input
Server configuration	5	0	Secret-safe and optional states
Audio hash	2	0	SHA-256 vectors
AI guardrails	5	0	Normalization and claim rejection
Medicine Corner	2	0	Catalogue and bilingual safety/source data
Messaging	9	0	Source, media, retry, privacy, host controls
Total	**38**	**0**	Focused repository tests

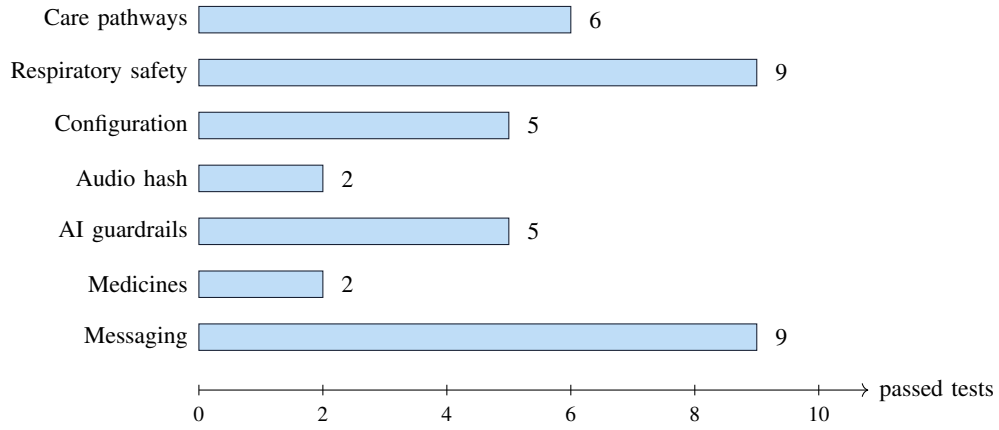


Fig. 11. Focused tests passed by suite; these are not clinical observations.

C. Feasibility Study

TABLE VIII
FEASIBILITY ASSESSMENT AND NEXT EVIDENCE GATE.

Dimension	Assessment	Evidence and next gate
Technical	Feasible prototype	Build/tests pass; validate live database, browser, and service paths
Operational	Partially feasible	Requires partners, verified facilities, staff identity, support ownership
Clinical	Not established	Requires expert review and prospective evaluation
Economic	Unknown	No deployment costing or cost-effectiveness study
Privacy/governance	Promising, incomplete	Formal legal, security, retention, and incident governance required
Scale	Plausible, untested	No load, availability, or nationwide field study

Technical feasibility is supported by the successful build, typed logic, validation, migrations, and focused checks. Operational feasibility is conditional on onboarding, facility acceptance, service ownership, and

reliable identity. Clinical feasibility is not established: questions, thresholds, translations, mappings, and messages require qualified review. Economic feasibility remains unknown because production infrastructure, support, monitoring, governance, and incentive costs have not been modelled.

D. Discussion

The closed loop links information to action: guidance can lead to a destination, referral can be independently completed, a missed visit can return to navigation, and programme signals can be derived without publishing private records. Deterministic rules trade predictive flexibility for auditability. At this stage, urgent precedence and language-neutral codes are easier to review and test than an opaque predictor.

The privacy design minimizes the signal rather than attempting to anonymize a complete record. Broad-area signals may support investigation or capacity planning, but cannot establish prevalence or a confirmed outbreak. Real deployment would need minimum cell sizes, disclosure control, temporal smoothing, governance, and epidemiological validation.

E. Challenges

Challenges include preserving simple language without weakening urgent instructions; maintaining equivalent translations; distinguishing overlapping pathways without diagnosis; verifying service capabilities beyond the TB-centred directory; preventing incentive distortion; maintaining cross-channel safety; governing audio data; and separating a polished prototype from an approved clinical system.

F. Limitations and Threats to Validity

The project lacks clinical, public-health, accessibility, security, legal, and regulatory approval. Content requires independent review. Facility coverage is incomplete. The staff PIN is a prototype control, credits are simulated, and human escalation does not alert a real team. Acoustic work is investigational and lacks the authorized data, calibration, subgroup analysis, and external validation needed for patient-facing triage.

The reported checks do not validate thresholds, translations, live RLS policies, facility accuracy, third-party availability, accessibility, load, or clinical outcomes. Browser suites, SQL regressions, live Supabase and optional-service checks, penetration testing, and human-subject studies were outside this report run. Controlled fixtures are regression evidence, not external-service validation.

V. Conclusion and Future Work

A. Conclusion

*First Signs*demonstrates a coherent approach to multi-syndrome care navigation in Bangladesh. It integrates deterministic safety guidance, destination finding, optional portable referral, independent attendance confirmation, barrier-aware follow-up, scoped incentives, and privacy-safe signals. The repository passed the focused static, build, and unit checks executed for this report. It remains a promising and testable prototype, not a diagnostic device or clinically validated service.

B. Future Scope

Priority work includes independent clinical, public-health, and language review; usability studies; verified pathway-specific facility data; accessibility and low-bandwidth testing; production identity, audit, monitoring, and incident response; disclosure controls for programme reporting; prospective measurement of referral acceptance and attendance; governed incentive and cost-effectiveness studies; interoperability; additional fully reviewed languages; load and penetration testing; and separately governed acoustic research using authorized data and preregistered evaluation.

C. Artifact Availability

The source is `docs/report/first-signs-ieee-report.tex`; references are in `docs/report/references.bib`; and repository-owned images are in `docs/images`. The document uses IEEEtran in one-column mode with an original project cover and report-style academic structure.

REFERENCES

[1] S. Ahmed, S. Biswas, T. Rahman, A. S. S. Adel, S. M. Z. Shafique, R. Tamanna, K. I. A. Chowdhury, and S. Banu, "Mixed-methods study to assess delay among patients with tuberculosis in an urban setting of bangladesh," *PLOS ONE*, 2025. [Online]. Available: https://doi.org/10.1371/journal.pone.0327348

[2] H. L. Semigran, J. A. Linder, C. Gidengil, and A. Mehrotra, "Evaluation of symptom checkers for self diagnosis and triage: Audit study," *BMJ*, vol. 351, p. h3480, 2015. [Online]. Available: https://doi.org/10.1136/bmj.h3480

[3] W. Wallace, C. Chan, S. Chidambaram, L. Hanna, F. M. Iqbal, A. Acharya, P. Normahani, H. Ashrafian, S. R. Markar, V. Sounderajah, and A. Darzi, "The diagnostic and triage accuracy of digital and online symptom checker tools: A systematic review," *npj Digital Medicine*, vol. 5, p. 118, 2022. [Online]. Available: https://doi.org/10.1038/s41746-022-00667-w

[4] E. Riboli-Sasco, A. El-Osta, A. Alaa, I. Webber, M. Karki, M. L. E. Asmar, K. Purohit, A. Painter, and B. Hayhoe, "Triage and diagnostic accuracy of online symptom checkers: Systematic review," *Journal of Medical Internet Research*, vol. 25, p. e43803, 2023. [Online]. Available: https://doi.org/10.2196/43803

[5] World Health Organization, *WHO Guideline: Recommendations on Digital Interventions for Health System Strengthening*. Geneva, Switzerland: World Health Organization, 2019. [Online]. Available: https://www.who.int/publications/i/item/9789241550505

[6] ——, "SMART guidelines," WHO Digital Health and Innovation, 2026, accessed 19 October 2026. [Online]. Available: https://www.who.int/teams/digital-health-and-innovation/smart-guidelines/

[7] A. Daftary, S. Satyanarayana, N. Jha, M. Singh, S. Mondal, C. Vadnais, and M. Pai, "Can community pharmacists improve tuberculosis case finding? a mixed methods intervention study in india," *BMJ Global Health*, vol. 4, no. 3, p. e001417, 2019. [Online]. Available: https://doi.org/10.1136/bmjgh-2019-001417

[8] T. Garg, V. Panibatla, J. P. Carel, A. Shanta, M. Bhardwaj, and M. Brouwer, "Can patient navigators help potential tb patients navigate the diagnostic and treatment pathways? an implementation research from india," *Tropical Medicine and Infectious Disease*, vol. 6, no. 4, p. 200, 2021. [Online]. Available: https://doi.org/10.3390/tropicalmed6040200

[9] M. A. B. Shiddik, F. Z. Toshi, S. Yesmin, and M. S. Rahman, "District-level dengue early warning prediction system in bangladesh using hybrid explainable AI and bayesian deep learning," *Tropical Medicine and Infectious Disease*, vol. 11, no. 3, p. 73, 2026. [Online]. Available: https://doi.org/10.3390/tropicalmed11030073

[10] S. M. Ahmed, R. Haque, U. Haque, and A. Hossain, "Knowledge on the transmission, prevention and treatment of malaria among two endemic populations of bangladesh and their health-seeking behaviour," *Malaria Journal*, vol. 8, p. 173, 2009. [Online]. Available: https://doi.org/10.1186/1475-2875-8-173

[11] C. M. George, S. Monira, F. Zohura, E. D. Thomas *et al.*, "Effects of a water, sanitation, and hygiene mobile health program on diarrhea and child growth in bangladesh: A cluster-randomized controlled trial of the cholera hospital-based intervention for 7 days (CHoBI7) mobile health program," *Clinical Infectious Diseases*, vol. 73, no. 9, pp. e2560–e2568, 2021. [Online]. Available: https://doi.org/10.1093/cid/ciaa754

[12] M. L. M. D. Souza, G. A. Lopes, A. C. Branco, J. K. Fairley, and L. A. D. O. Fraga, "Leprosy screening based on artificial intelligence: Development of a cross-platform app," *JMIR mHealth and uHealth*, vol. 9, no. 4, p. e23718, 2021. [Online]. Available: https://doi.org/10.2196/23718

[13] M. Zaman, R. A. Hridhee, R. A. Bhuiyan, C. A. Gomes, M. M. Rahman, S. M. S. Islam, F. Sarker, and K. A. Mamun, "Efficacy of using a digital health intervention model using community health workers for primary health services in bangladesh: A repeated cross-sectional observational study," *BMC Public Health*, vol. 25, p. 1833, 2025. [Online]. Available: https://doi.org/10.1186/s12889-025-22770-9

[14] S. Baur, Z. Nabulsi, W.-H. Weng, J. Garrison, L. Blankemeier, S. Fishman, C. Chen, S. Kakarmath, M. Maimbolwa, N. Sanjase, B. Shuma, Y. Matias, G. S. Corrado, S. Patel, S. Shetty, S. Prabhakara, M. Muyoyeta, and D. Ardila, "HeAR—health acoustic representations," *arXiv preprint arXiv:2403.02522*, 2024. [Online]. Available: https://arxiv.org/abs/2403.02522